APS110H1f2026 - September 28, 2026

September 25, 2026 12:34 PM

- Properties of Matter(s) - today again (and also for this week as well again) - also - "I Need More Structure in My Life"
  - Calculation
  - Radius
  - Nano structure - Bonding
  - Atomic Packing Factor - maybe

Radius - maybe today - Copied it to here.....

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r = radius.



2D

pm = picometre



d - diameter "not APS110"



3D

$$\text{volume} = \frac{4}{3}\pi r^3$$

π 3.141592650 5 1

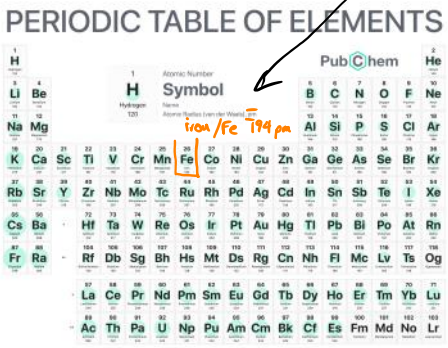
π 3.14159 or 2

π 3.14160 or 3

only Fe/iron 194 pm

$$= \frac{4}{3} \cdot 3.14160 \cdot (194\ \text{pm})^3$$

$$= 2437.88 \text{ volume re pm.}$$



1 pm = 0.001 nm

1000 pm = 1 nm

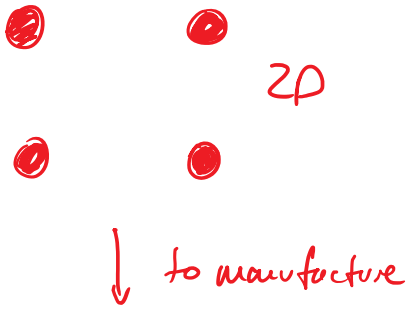
1 pm = $1e^{-10}$ cm = 0.00000000001 cm

1 cm = $1e^{10}$ pm = 10000000000 pm

Bonding

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just as an example.



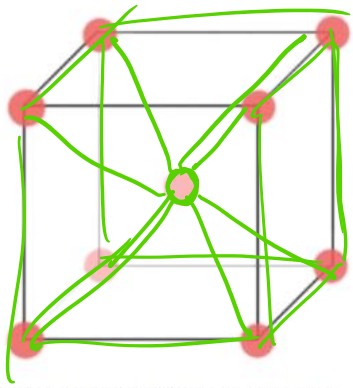
2D

to manufacture



2D

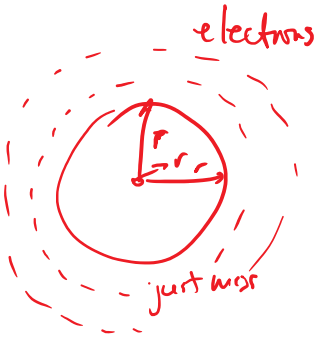
more density



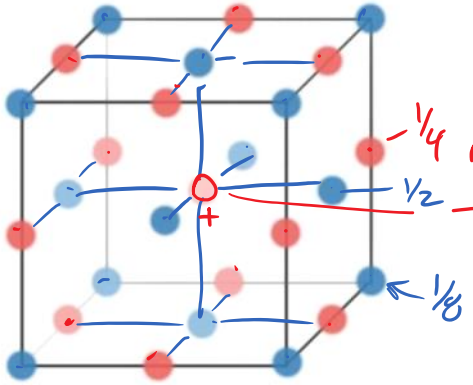
metal only.

bonded.

Figure 4: The body centred cubic crystal structure.



Ceramics.



$12 \times 1/4 = 3$

$= 4$

I

$8 \times 1/8 = 1$

$6 \times 1/2 = 3$

$= 4$

Figure 2: The rock salt ceramic crystal structure.

just metals.

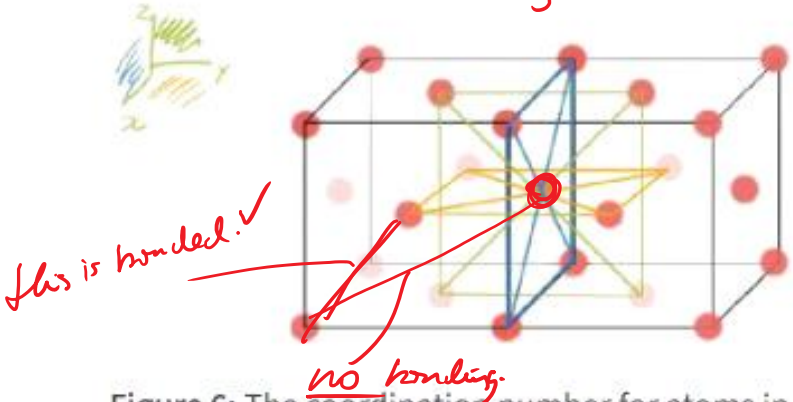
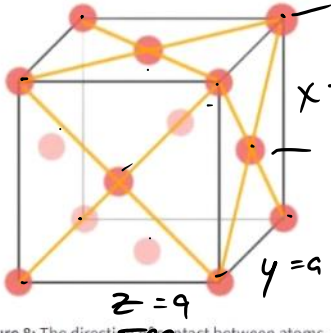


Figure 6: The coordination number for atoms in FCC. The right side face-centred atom makes contact with four atoms in the right side plane, with four atoms in the face-centred positions in the original unit cell, and with four atoms in teh face-centred positions in the unit cell to the right.

APF - atomic packing factor. = also density



1 + 3 = 4 = n

just Fe/iron

Figure 8: The direction of contact between atoms in the FCC unit cell is along the cube face diagonals, as illustrated by the orange lines.

$$APF = \frac{\text{volume spheres}}{\text{volume unit cell}}$$

$$= \frac{n \frac{4}{3} \pi R^3}{a^3}$$

$$= \frac{n \frac{4}{3} \pi R^3}{x \cdot y \cdot z}$$

$$= \frac{n \frac{4}{3} \pi R^3}{FCC^3} = \frac{n \frac{4}{3} \pi R^3}{(2\sqrt{2}R)^3}$$

or angle

FCC - face centred cubic

$a_{FCC} = 2\sqrt{2}R$

$= 548.7096\ pm$

R - radius (of the atom)

π - 3.14159

$$APF = \frac{4 \cdot \frac{4}{3} \cdot 3.14159 \cdot (194\ pm)^3}{(548.7096\ pm)^3} = 668851.0380$$

density.